# Monte Carlo Pricing – Convergence and Performance Summary

**Master 2 – Financial Engineering - Quantitative Finance Track**

Paris Dauphine University - PSL

**Students:** Renault Léo - Théo Verdelhan

**Course:** Monte Carlo Methods for Derivative Pricing

**Date:** March 1, 2026

---

## 1. Objective

The objective of this project is to implement a Monte Carlo pricing engine for European and American options under the risk-neutral GBM framework, and to analyze:

- Convergence of price and Greeks,
- Variance reduction techniques,
- Scalar vs vectorized performance,
- Early exercise behavior,
- Detection and explanation of anomalous prices.

## 2. Methodology (Summary)

The underlying follows:

$$S_{t+\Delta t} = S_t \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}Z\right)$$

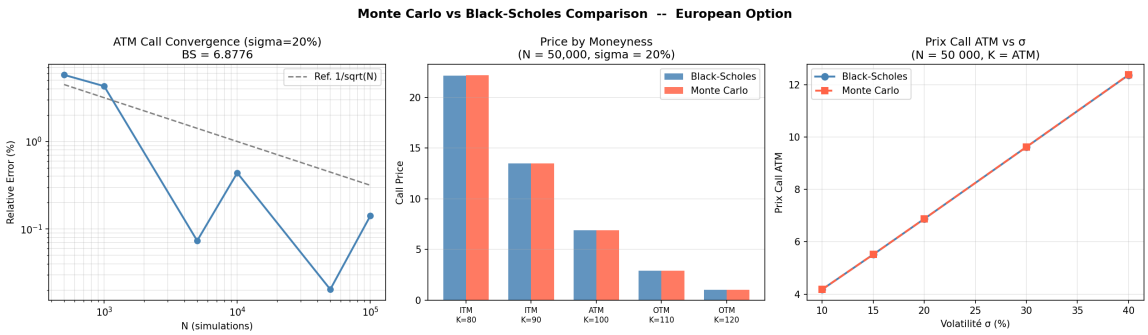
European pricing uses:

$$V_0 = e^{-rT}\mathbb{E}[h(S_T)]$$

American pricing uses the Longstaff–Schwartz regression method (quadratic basis), benchmarked against a trinomial tree.

Variance reduction is implemented via antithetic variables. Greeks are computed using central bump-and-reprice with Common Random Numbers.

# 3. European Price Convergence

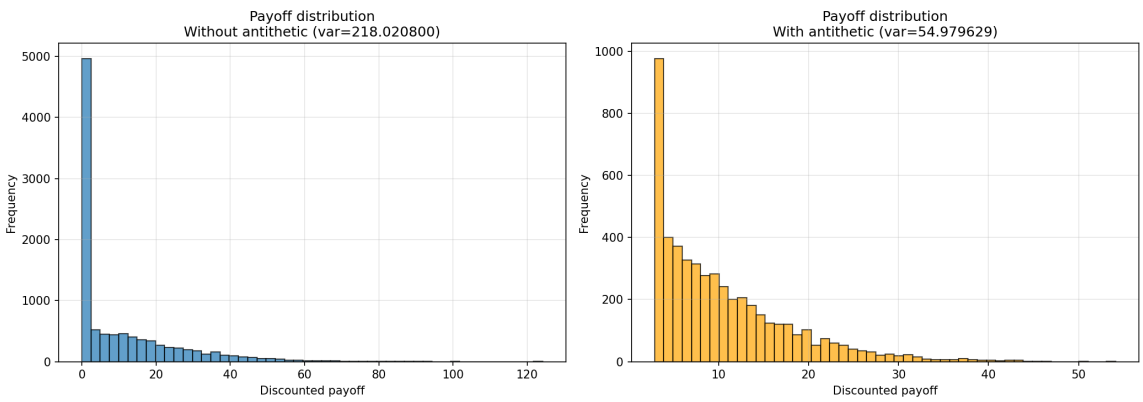



The Monte Carlo estimator converges toward the analytical Black–Scholes price as the number of paths increases.

The log-log slope of the standard error is approximately $-0.5$, confirming the theoretical $O(N^{-1/2})$ convergence predicted by the Central Limit Theorem.

Confidence intervals shrink proportionally to $1/\sqrt{N}$, and the benchmark lies within the confidence band for sufficiently large $N$.

# 4. Variance Reduction — Antithetic Variables

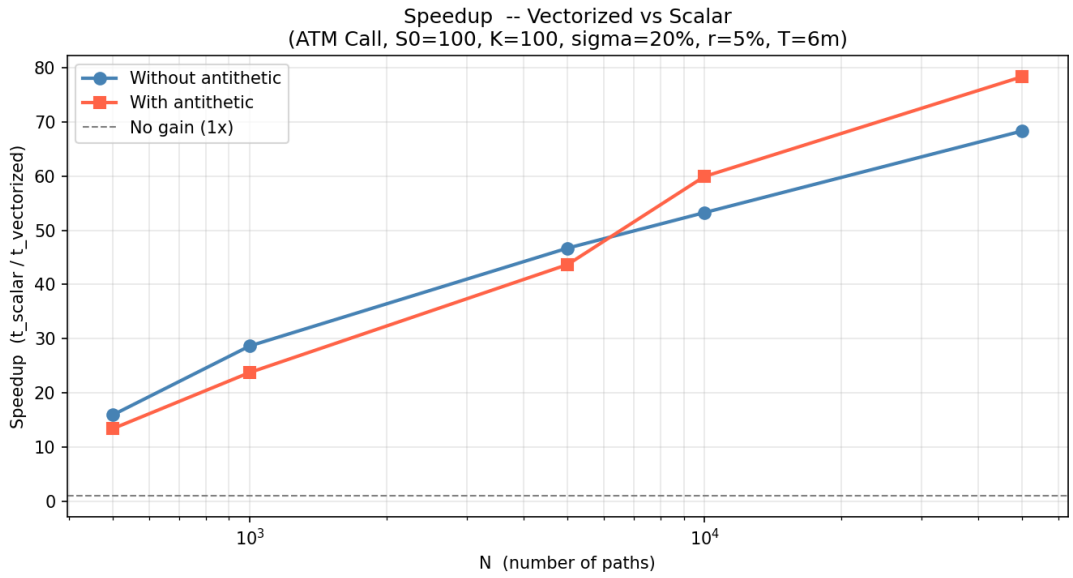



Antithetic pairing significantly reduces estimator variance while preserving unbiasedness.

The empirical slope remains close to $-1/2$, but the variance constant is reduced, leading to precision gains approaching a factor of two.

This confirms the effectiveness of negative correlation between paired paths.

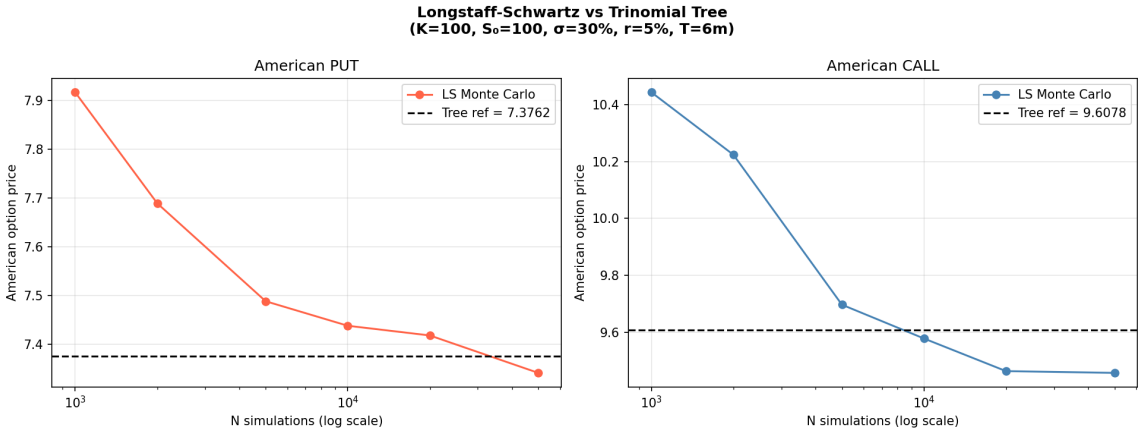
## 5. Scalar vs Vectorized Performance




Both implementations scale as $O(NM)$. However, the vectorized implementation leverages NumPy's C-level loops and contiguous memory layout, reducing runtime by more than 20× and up to 70× for large simulations.

Numerical outputs are identical up to floating-point precision.

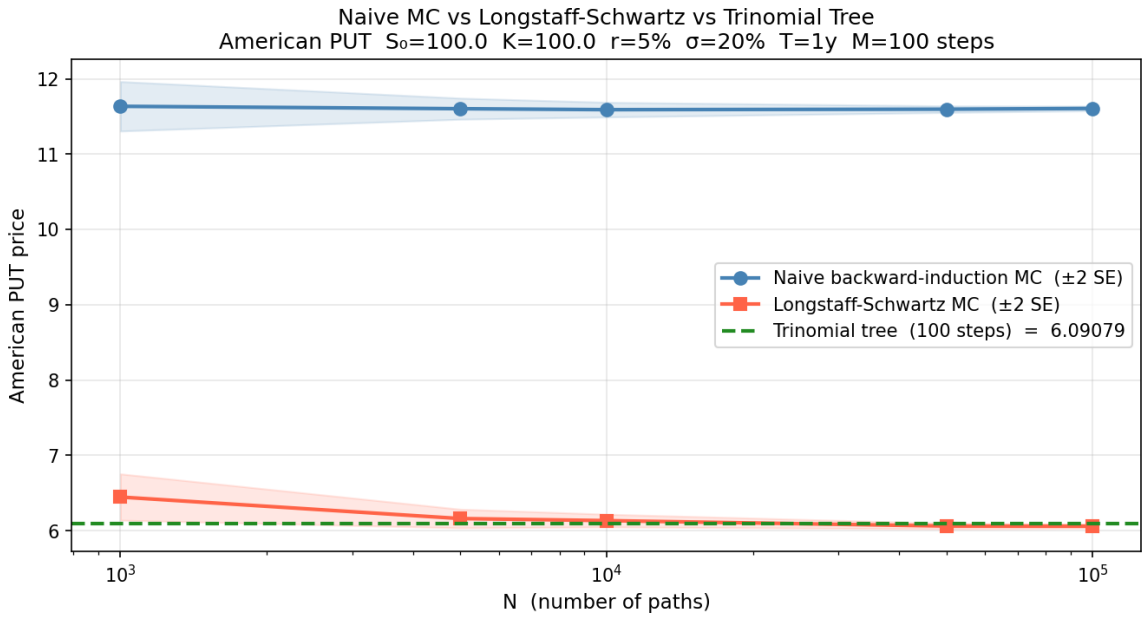
## 6. American Pricing and Early Exercise




Longstaff–Schwartz prices converge toward the trinomial tree benchmark as $N$ increases.

The regression-based continuation value correctly captures optimal stopping behavior.

### Naive Monte Carlo Bias



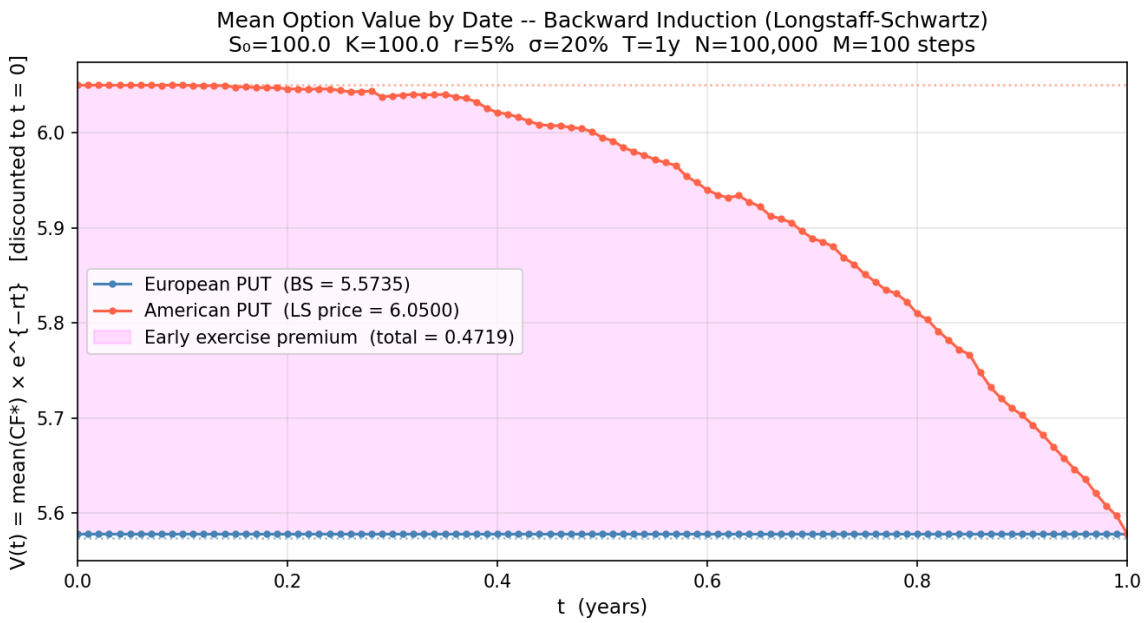

The naive backward-induction Monte Carlo produces a significantly inflated price ( 11.6 vs 6.1). This is not statistical noise but structural bias:

$$\mathbb{E}[\max(h, X)] > \max(h, \mathbb{E}[X])$$

The naive method uses realized future information, violating the adapted filtration of the stopping problem.

The bias does not disappear as $N \to \infty$.

## 7. Mean Option Value by Date

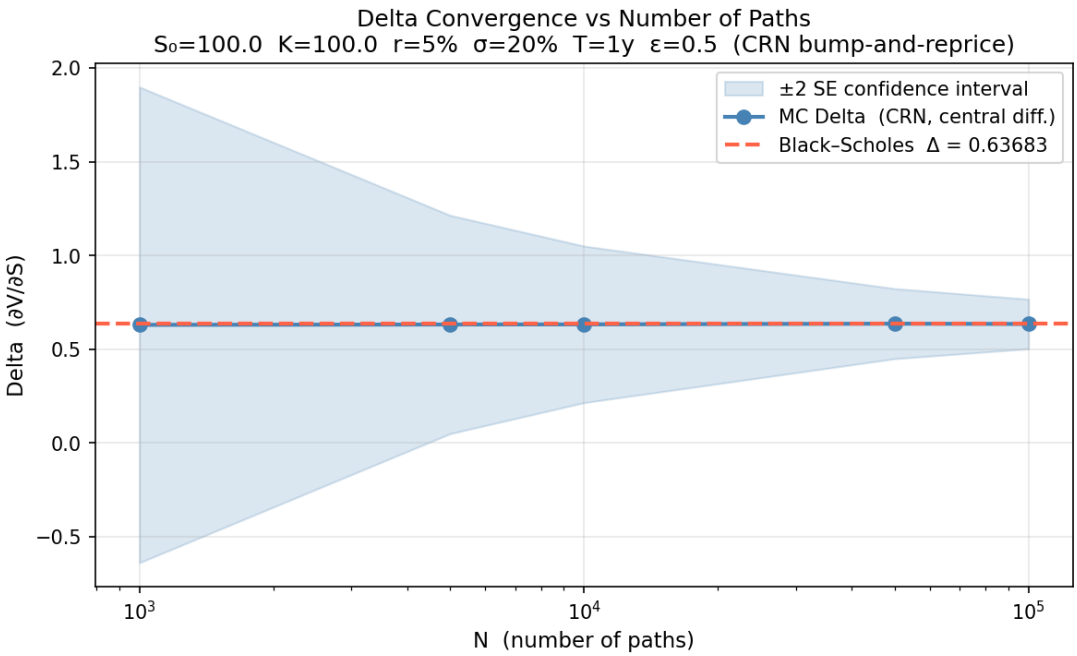


The discounted European option value remains approximately constant, consistent with the martingale property under $\mathbb{Q}$.

The discounted American value decreases backward in time, forming a supermartingale due to early exercise.

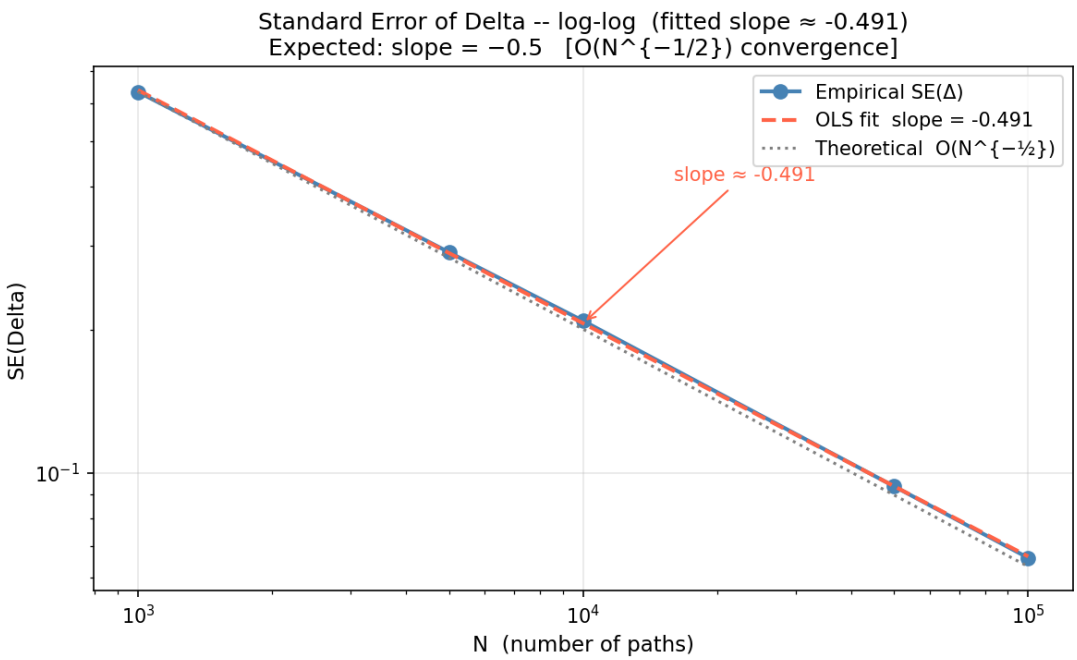
The shaded region represents the early exercise premium.

## 8. Greeks Convergence — Delta




Delta converges toward the analytical Black–Scholes value as $N$ increases.

Common Random Numbers significantly reduce variance in the finite-difference estimator.




The standard error of Delta follows the same $O(N^{-1/2})$ decay rate, with empirical slope $-0.49$.

## 9. Conclusion

The numerical experiments confirm:

- Theoretical $O(N^{-1/2})$ convergence of price and Delta.
- Effective variance reduction via antithetic variables.
- Major computational gains from vectorization.
- Structural upward bias of naive American Monte Carlo.
- Correct early-exercise structure under Longstaff–Schwartz.